A
report
on
Project Based Learning
on

Title : CLI-Based Linux Server Administration and Automated Backup & Recovery System

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1. Problem Statement:

Linux systems require regular administration of users, groups, file permissions, services, and backups. Performing these tasks manually through separate terminal commands can be repetitive and may lead to mistakes.

This project addresses the need for a simple command-line tool that brings common Linux administration tasks together. It uses Bash scripts to manage accounts and groups, configure a shared directory, enable secure file transfer, and back up and restore shared data.

2. Abstract:

This project presents a CLI-Based Linux Server Administration and Automated Backup & Recovery System developed using Bash shell scripting. A central menu provides access to User & Group Management, File Management & Secure Sharing, Backup & Recovery, and System Information. The user module creates and deletes accounts, manages groups and membership, locks or unlocks passwords, and lists users and groups. The file module configures a group-controlled shared directory, displays its permissions and contents, tests user access, and enables or checks SSH/SFTP through OpenSSH. The backup module creates timestamped compressed archives, lists and restores backups, maintains logs, applies retention, and supports daily scheduling through cron. The project uses standard Linux utilities through separate Bash modules.

3. Introduction:

Linux provides command-line utilities for managing accounts, permissions, services, and data. A menu-driven administration tool can make common operations easier to access while continuing to use the standard Linux utilities underneath. This project organizes the operations into modules and uses shared configuration and helper functions to keep the application consistent.

4. Objective(s):

- To develop a menu-driven Linux administration tool using Bash shell scripting.
- To manage local users and groups and configure a controlled shared directory.
- To test shared-directory access and enable SSH/SFTP services.
- To create, list, and restore compressed backups and automate backup scheduling.
- To display basic system information and record administrative activity.

5. Methodology:

Project Environment and Tools:
The project is written in Bash and uses standard Linux command-line utilities. Ubuntu Desktop is the team's working environment. The application runs as a normal user and requests sudo for individual administrative operations.

System Architecture:
The main.sh file displays the menu and starts the selected module. Feature scripts are kept under modules/, common functions are stored in lib/common.sh, and shared paths and settings are defined in config/settings.conf.

User & Group Management:
The user module creates and deletes local accounts, creates groups, adds users to supplementary groups, locks or unlocks passwords, and lists users and groups. It uses commands such as useradd, userdel,

groupadd, usermod, passwd, and getent.

File Management & Secure Sharing:
The file module configures /srv/linux-admin/shared for group linuxshare with owner root and permission mode 3770. It can add users to the group, show directory permissions and contents, test read/write/traverse access, and enable or check SSH/SFTP through OpenSSH.

Backup Creation:
The backup module creates timestamped compressed .tar.gz archives of the configured shared directory in /var/backups/linux-admin. The helper script records backup activity and applies the configured retention period after a successful backup.

Backup Recovery:
The restore option lists available archives and lets the administrator select one. The helper validates the archive, creates a safety backup, extracts to a temporary staging directory, and restores archived files using a merge approach that retains unrelated extra files.

Automatic Backup Scheduling:
The project can install a cron job using the default schedule 0 2 * * *, meaning every day at 02:00 local machine time. The installer places required files in a root-owned location and creates the schedule under /etc/cron.d.

System Information and Testing:
The system-information module displays host, network, uptime, memory, disk, and SSH details. Testing uses dedicated test accounts and sample files to check account operations, directory permissions, SSH status, backup creation, restoration, cron setup, and script syntax.

6. Results and Discussion:

Output screenshots, graphs, or tables:

The project output is presented through the main menu and the interactive modules. The screenshots show the expected menu structure and sample terminal operations for managing users, groups, shared directories, SSH/SFTP, and backup recovery. The displayed screens are illustrative examples of the expected output and should be replaced with screenshots captured from the team's final working Ubuntu system after testing.

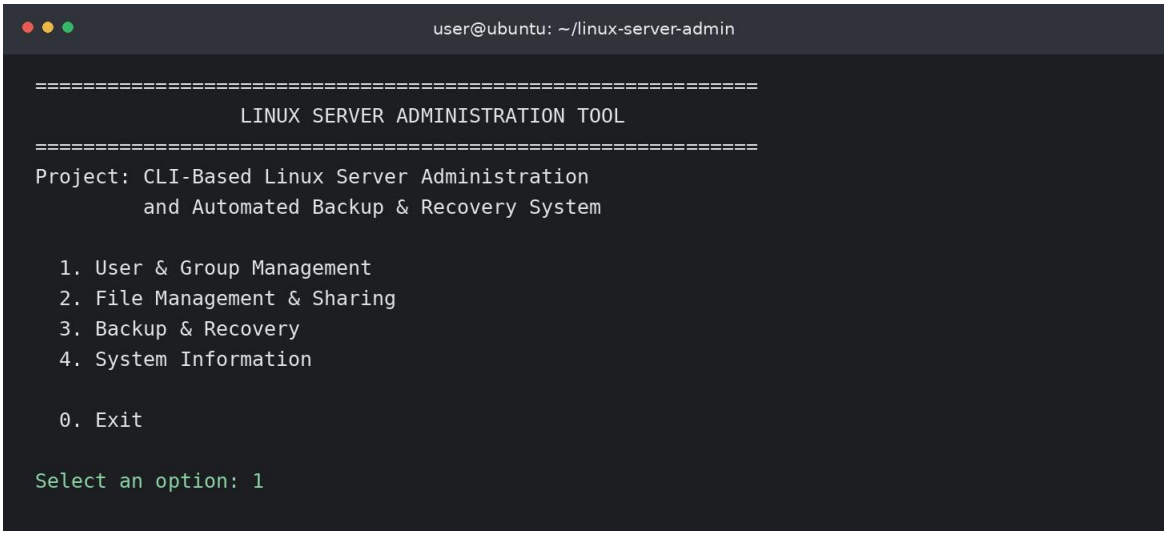


Fig1: Main menu of Linux Server Administration Tool

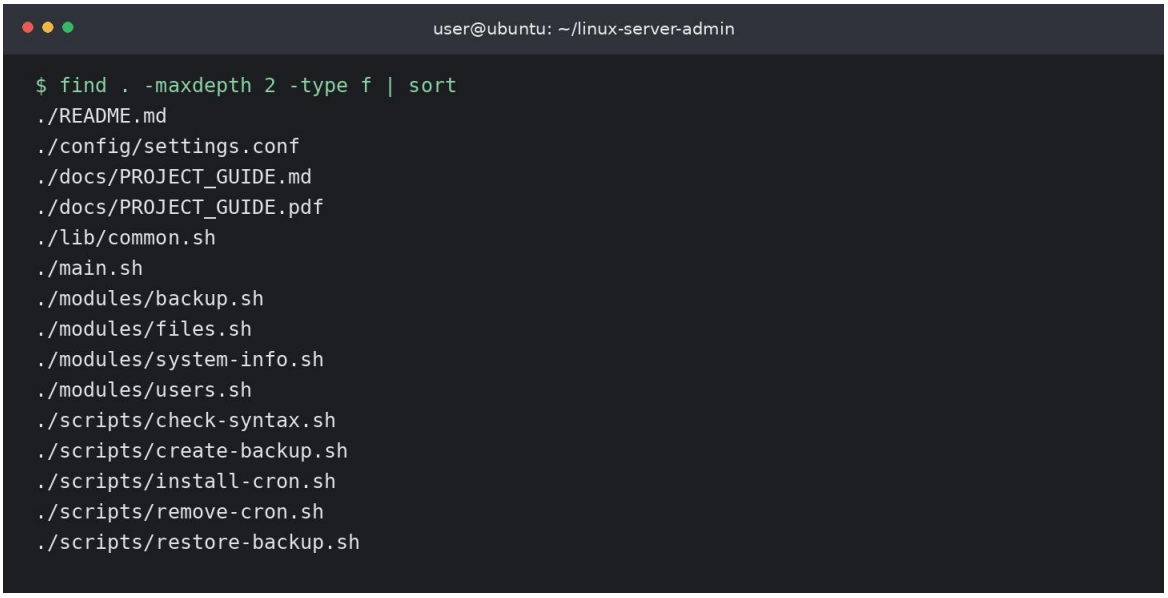


Fig2: Project directory structure

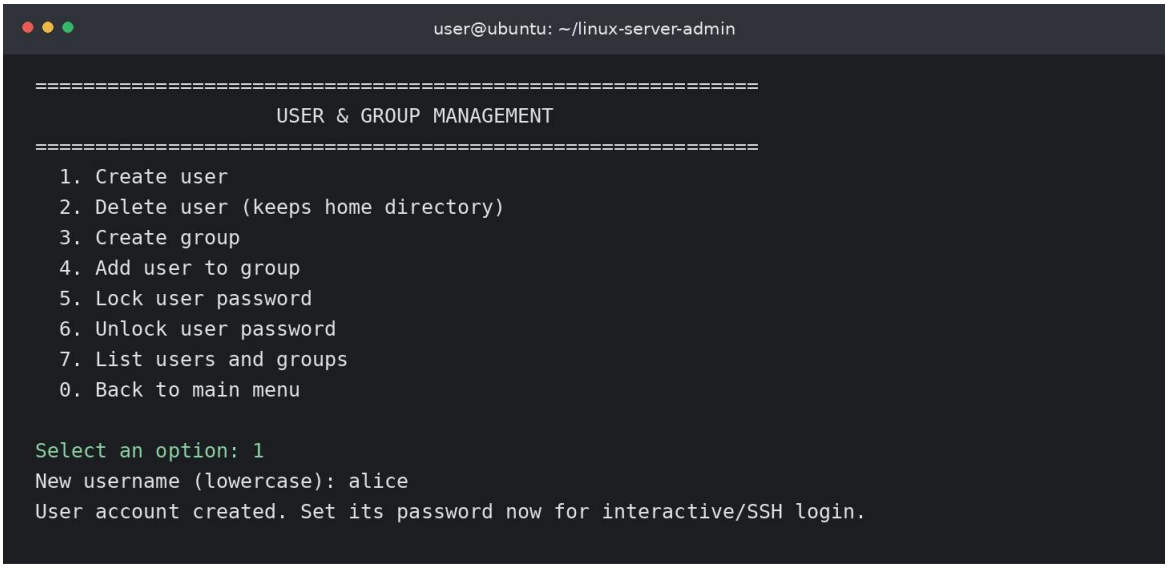


Fig3: User & Group Management menu and create-user option

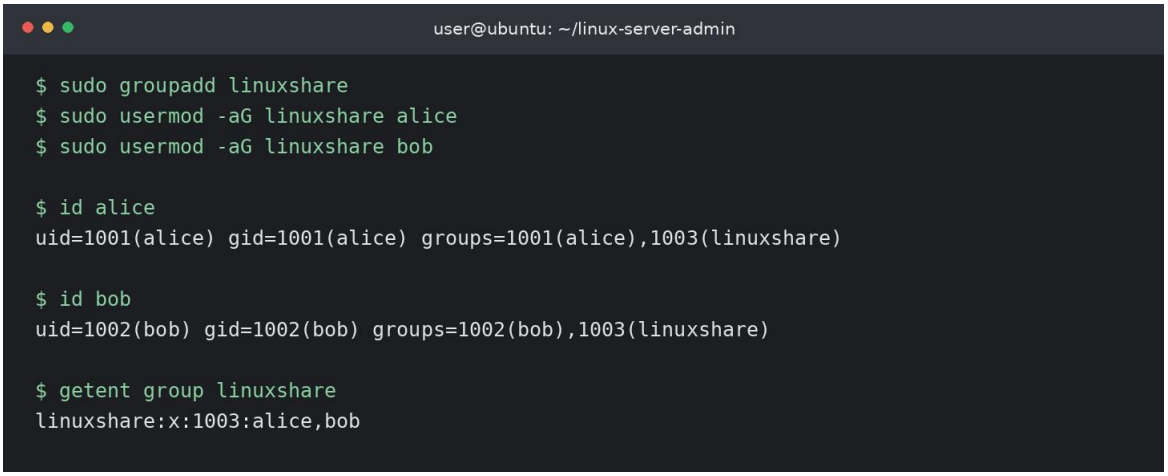


Fig4: User and group membership verification using Linux commands

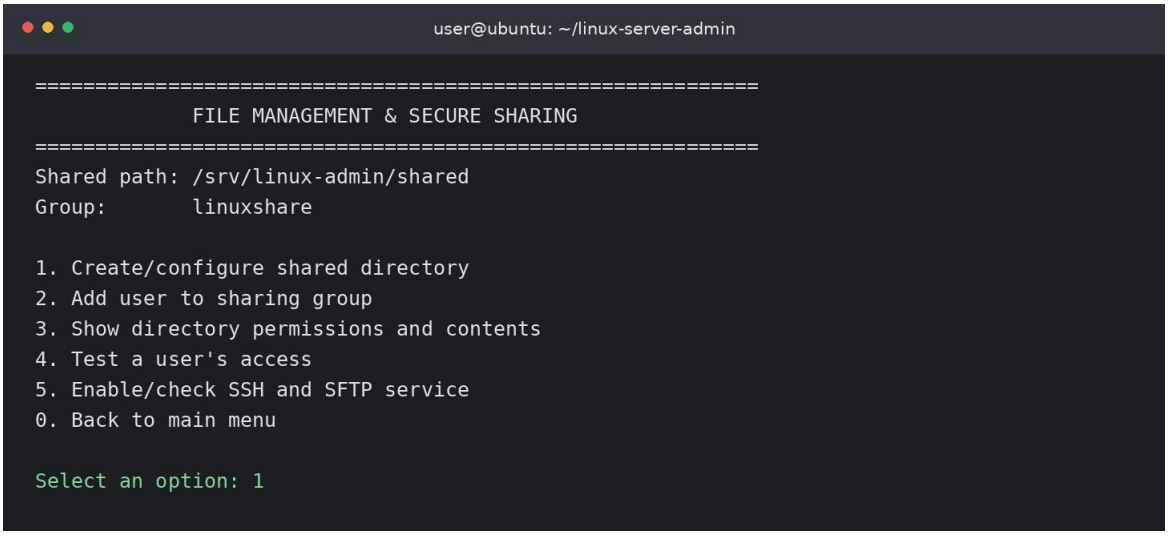


Fig5: File Management & Secure Sharing menu

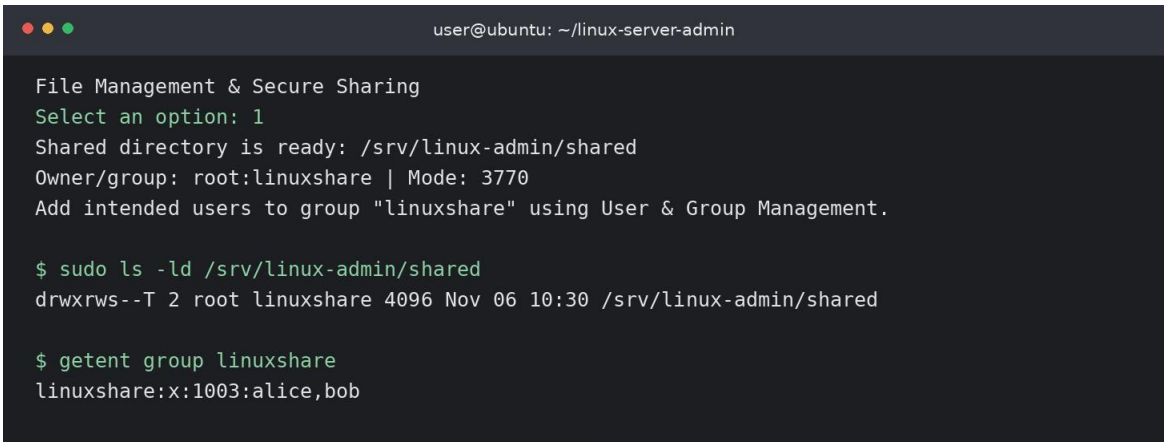


Fig6: Shared directory setup and ownership verification

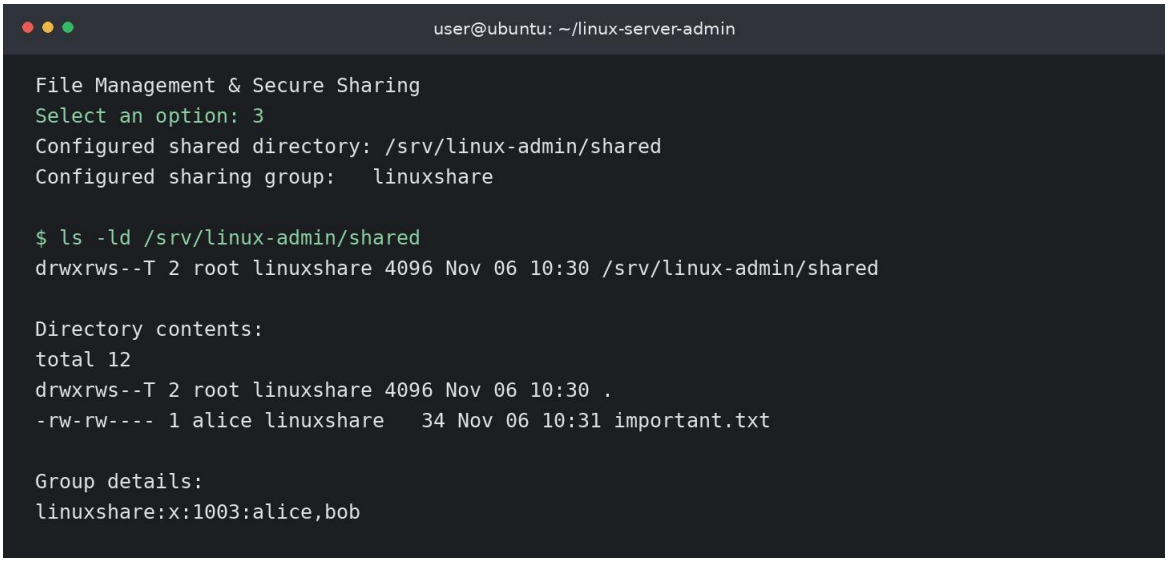


Fig7: Shared directory permissions and contents

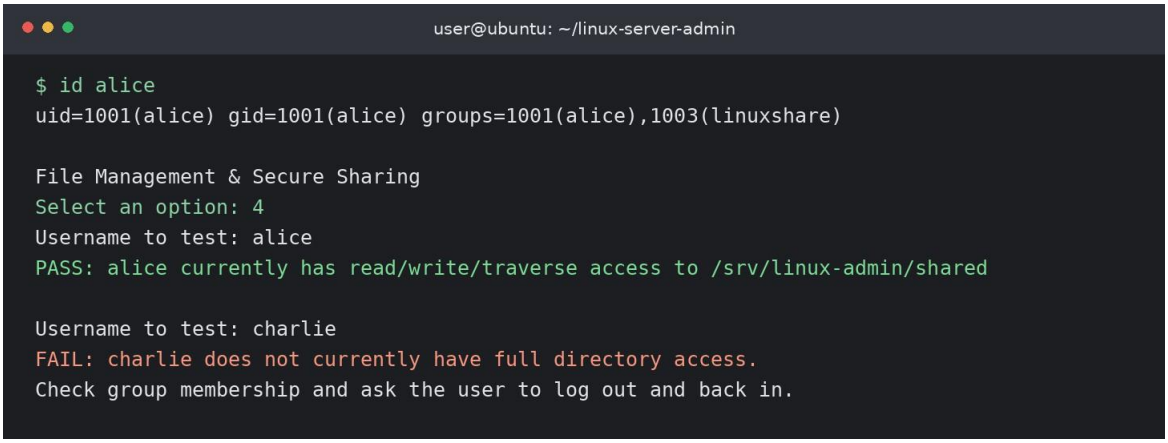


Fig8: Access test for a group member and a non-member

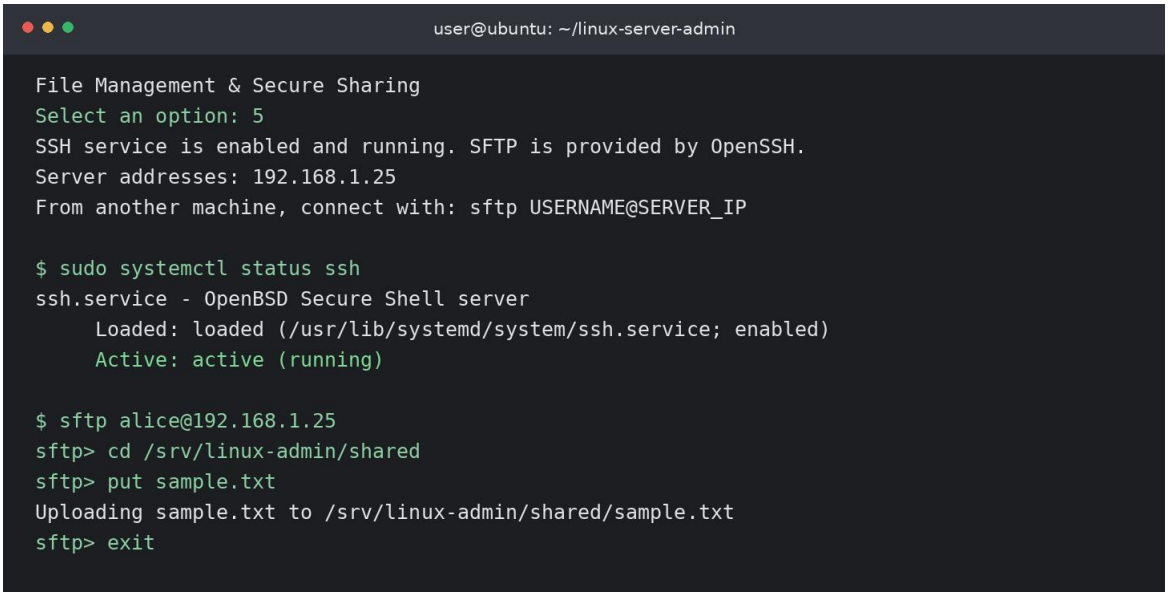


Fig9: SSH service status and SFTP file transfer

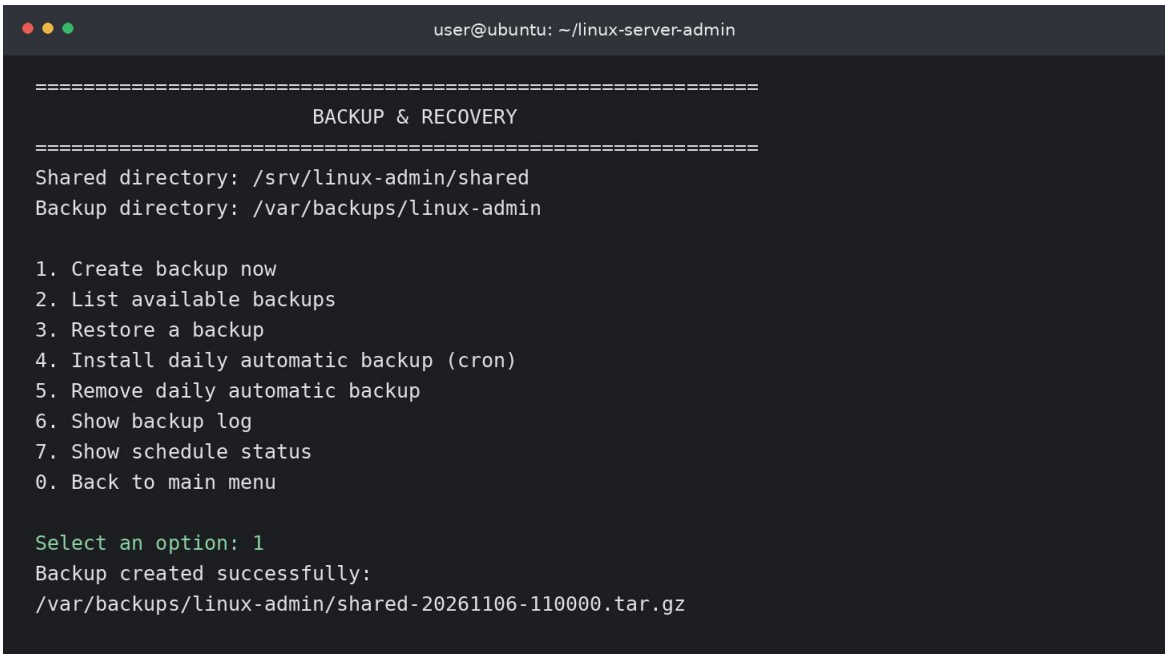


Fig10: Backup & Recovery menu and backup creation



Fig11: Compressed backup archive listing and contents

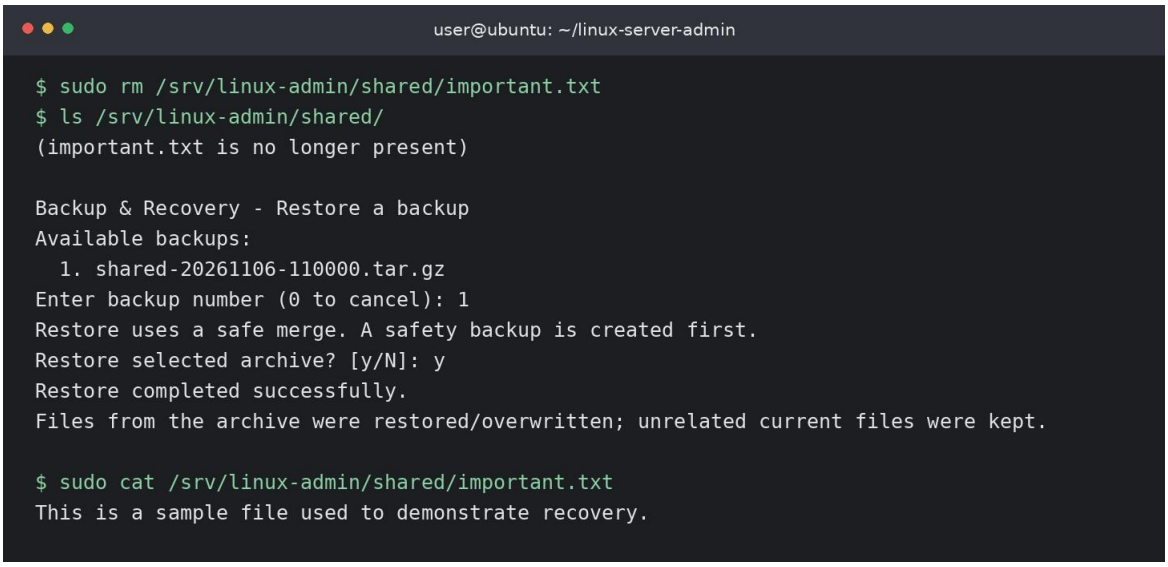


Fig12: Restoration of a deleted sample file from backup

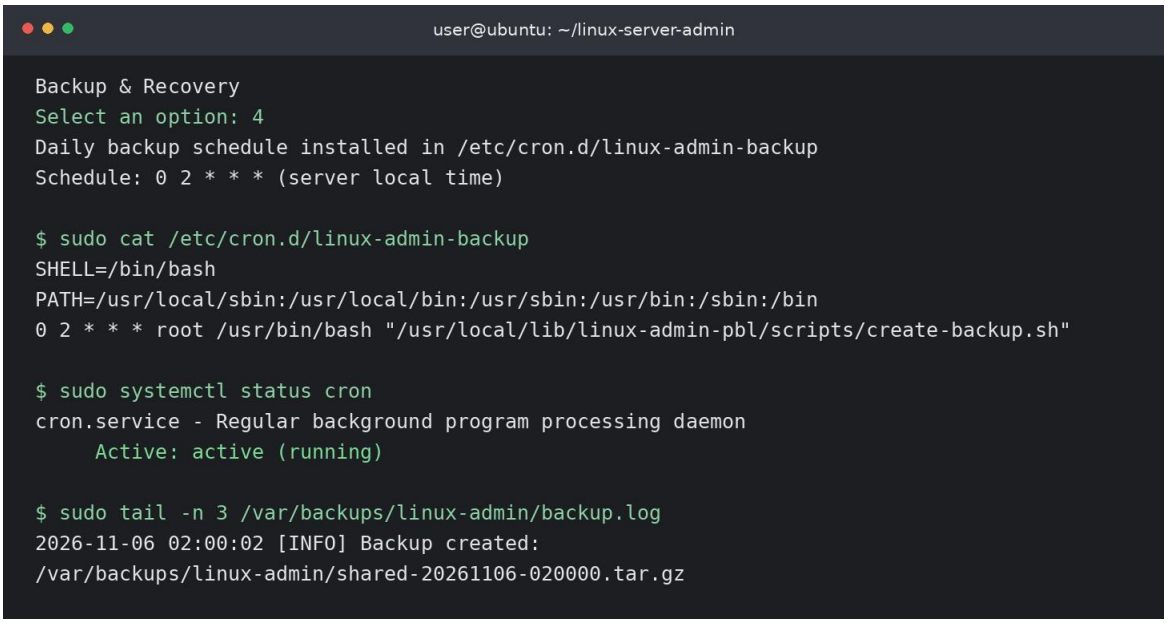


Fig13: Daily cron schedule and backup log

Observations:

The project separates common administration operations into three primary working modules and a system-information module. User and group management provides a way to create accounts and

assign group membership. The shared-directory configuration uses Linux ownership and permission settings to control which users can access shared files. The access-check option is dependent on the shared directory being created and on the selected user having the appropriate group membership.

The file-sharing module uses OpenSSH to provide SSH and SFTP functionality rather than implementing a separate file-transfer protocol. The backup module stores compressed archives with timestamps so that different backup versions can be identified. The restore helper creates a safety copy before restoring archived content and follows a merge-based approach. The automatic backup option uses cron, while the backup log provides a record of backup operations.

Interpretation of Results:

The project demonstrates how Bash scripting can combine existing Linux utilities into a single menu-driven administration application. The user and group module handles local account operations, while the file module applies group-based directory permissions and provides an access test. The backup module supports the protection and recovery of the configured shared directory. These functions demonstrate basic system administration tasks in a structured and repeatable manner.

The current project is intended as an educational implementation and not as a production-grade disaster-recovery platform. Backups are stored on the same machine by default, so a failure of the VM disk may affect both the original files and their backups. The project backs up only the configured shared directory; it does not create a full operating-system image or back up all home directories and system configuration files. Remote backup storage, advanced ACLs, and continuous monitoring are outside the current scope.

7. Learning Outcomes:

- Understanding of Linux user and group administration.
- Practical knowledge of file ownership and permission settings using chmod and chown.
- Understanding of shared-directory access control using Linux groups, setgid, and sticky bit.
- Experience with Bash functions, shell scripts, menus, and case statements.
- Practical exposure to SSH service management and SFTP file transfer using OpenSSH.
- Understanding of compressed backups and file restoration using tar and related Linux utilities.
- Knowledge of cron scheduling, configuration files, and log management.

8. Conclusion:

The CLI-Based Linux Server Administration and Automated Backup & Recovery System brings common Linux administration operations together in a menu-driven Bash application. It provides options for managing local users and groups, configuring a group-controlled shared directory, checking access permissions, enabling SSH/SFTP, creating compressed backups, and restoring selected data. The cron feature supports scheduled daily backups, while the shared configuration and helper functions keep the modules organized.

The project provides practical learning in Linux system administration, Bash scripting, access control, secure file transfer, and backup management. It can be extended in the future by storing backups on a separate machine or storage device, improving backup verification, and adding further access-

control features. The current scope remains focused on basic administrative functions that can be demonstrated and explained clearly.

9. References:

1. GNU Project. Bash Reference Manual. Available at: https://www.gnu.org/software/bash/manual/
2. Canonical. Ubuntu Server Documentation. Available at: https://documentation.ubuntu.com/server/
3. Ubuntu Manpage Repository. Linux command references: useradd, userdel, groupadd, usermod, passwd, chmod, chown, tar, and cron. Available at: https://manpages.ubuntu.com/
4. OpenSSH Project. OpenSSH Manual Pages. Available at: https://www.openssh.com/manual.html
5. GNU Project. GNU tar Manual. Available at: https://www.gnu.org/software/tar/manual/
6. Project source code: main.sh, lib/common.sh, config/settings.conf, modules/users.sh, modules/files.sh, modules/backup.sh, modules/system-info.sh, and scripts/.
7. Linux Server Administration PBL - Starter Implementation Guide supplied with the project.